

TrES-3b

R-0138

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230508 · created 2026-07-16T07:11:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460072.8906 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.022
Transit depth (Rp/Rs)^2	2.08 +/- 0.64 [%]
Orbital Inclination (inc)	83.0 +/- 0.92
Airmass coefficient 1 (a1)	1.115 +/- 0.015
Airmass coefficient 2 (a2)	-0.103 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.32 %
Best Comparison Star	None
Optimal Aperture	3.95
Optimal Annulus	7.73
Transit Duration (day)	0.0614 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.144 $\pm$ 0.022 vs 0.16309 (deviation 0.81 σ)
Duration fit vs published	0.0614 d vs 0.0567 d (deviation 0.63 σ)
Mid-time O-C	-3.9 min
Depth SNR	6.1
Residual scatter	1.32 %

Light curve

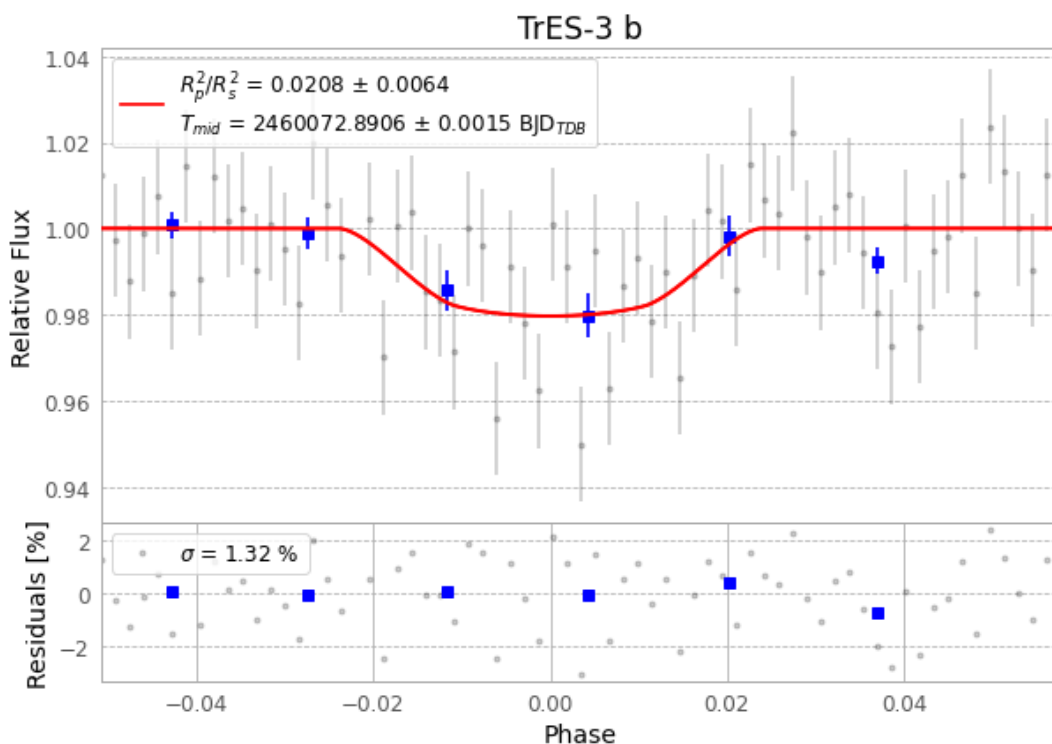


Figure 1 — FinalLightCurve_TrES-3 b_2023-05-08.png (SHA-256 7dee0b4e399e5ffc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 83a5cfd3fa32e722...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

69 FITS frames (45 MB), TRES-3230508074215.FITS → TRES-3230508110615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230508.log (e727fc787263...), FinalLightCurve_TrES-3 b_2023-05-08.png (7dee0b4e399e...), FinalLightCurve_TrES-3 b_2023-05-08.csv (4c4e5071e348...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 07:11Z.